

	Work Instruction	Doc. Revision	02
	AXI V810 X-Axis Limit Sensor Adjustment Procedure	Effective Date	20 Oct 2016

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	19 Aug 2013	First Release	ONG KEAN SHENG
01	04 Jun 2015	Second Release	MUHAMMED SHAMEEM
02	20 Oct 2016	Template Changes & Content Updates	ONG KEAN SHENG

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1. Close the V810 Software
2. Login to the Atom PC and enter the computer name as shown below in Figure 1 (S1 Series) and Figure 2. (S2 Series)

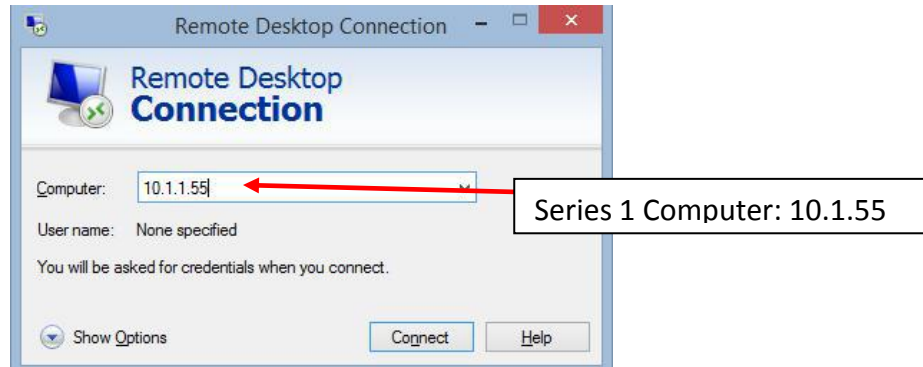


Figure 1 Series 1

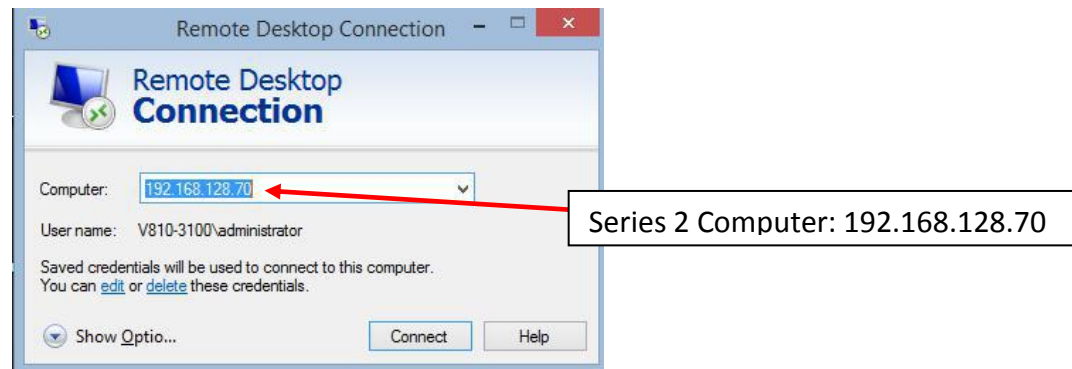


Figure 2 Series 2

3. Enter the password as "Please!" as shown in Figure 3.

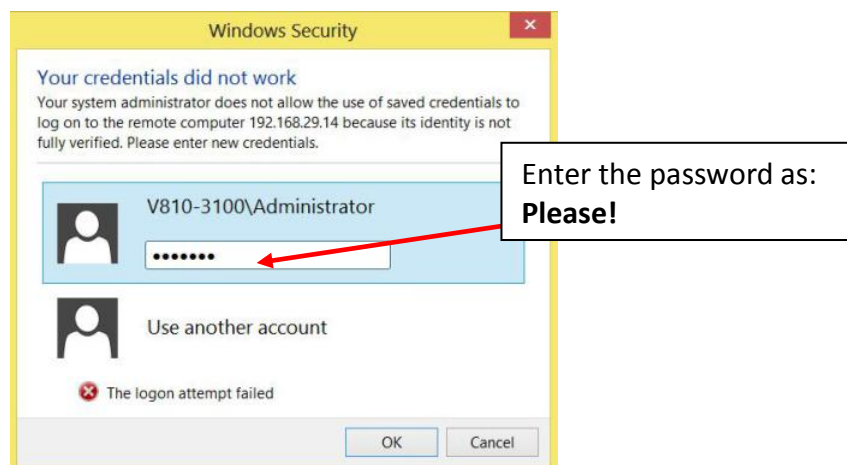


Figure 3 Password for Atom PC

4. Go to the following directory: C:\Program Files\AXI System\v810\REV\bin and open the MC_XMP_NT.EXE as shown in Figure 4.

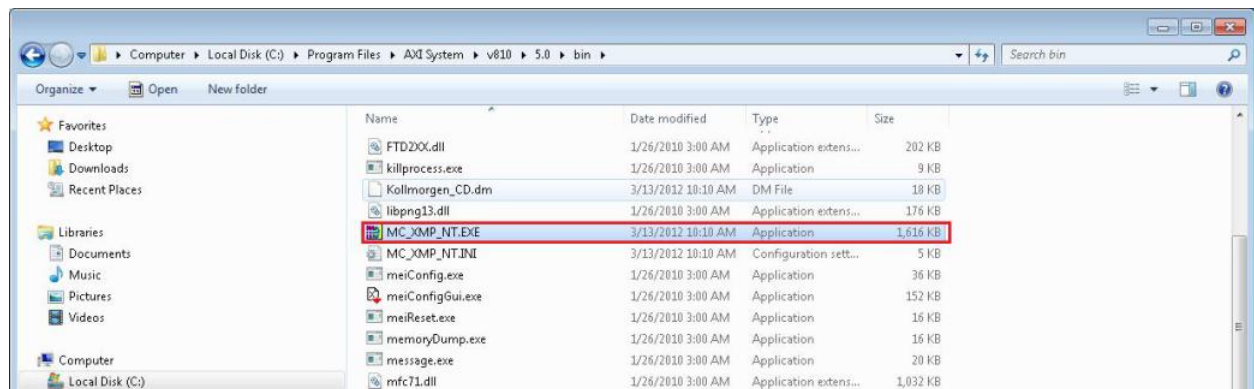


Figure 4 MC_XMP_NT.EXE

5. Ensure this 3 tab is open in the Motion Console tab as shown in Figure 5.

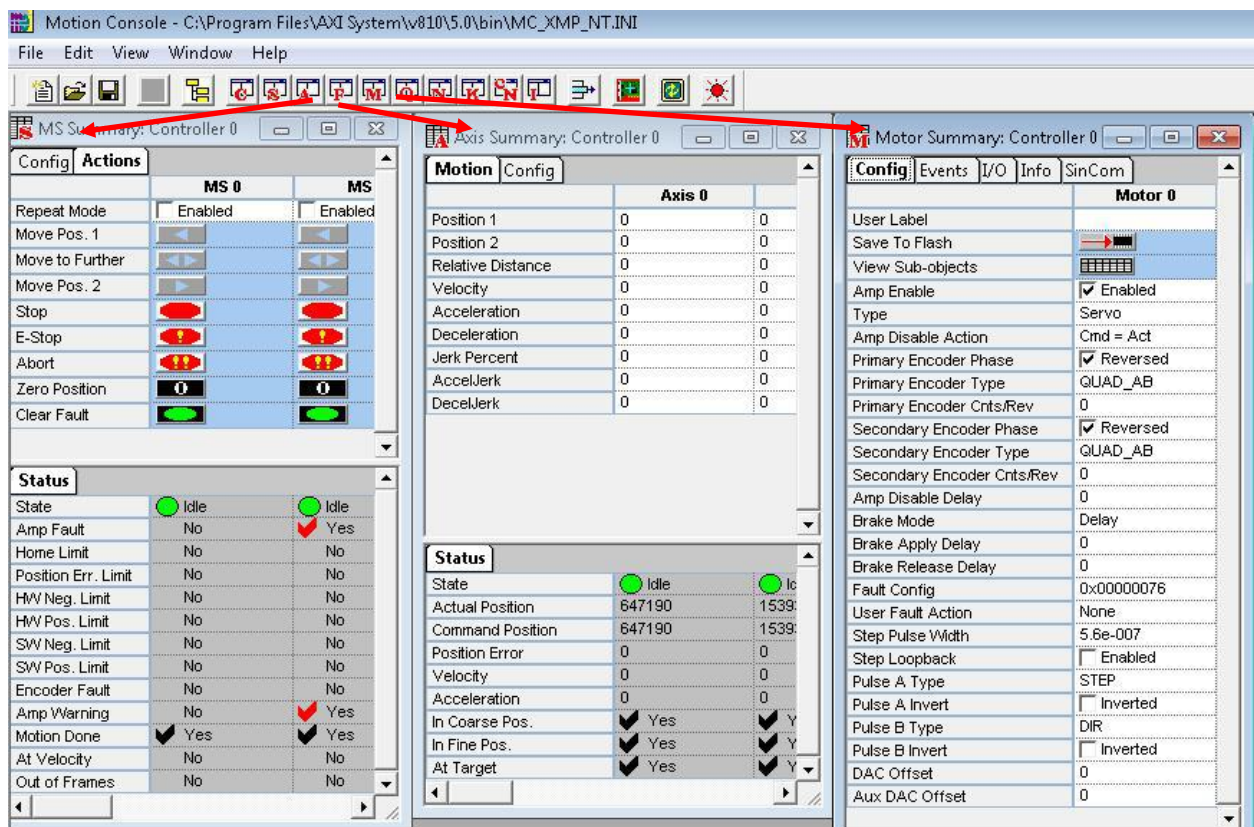


Figure 5 Motion Control Tab

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6. Close the X-ray source and open the front access door.
7. Push the stage towards the minimum hard stop (left outer barrier) as shown in Figure 6.

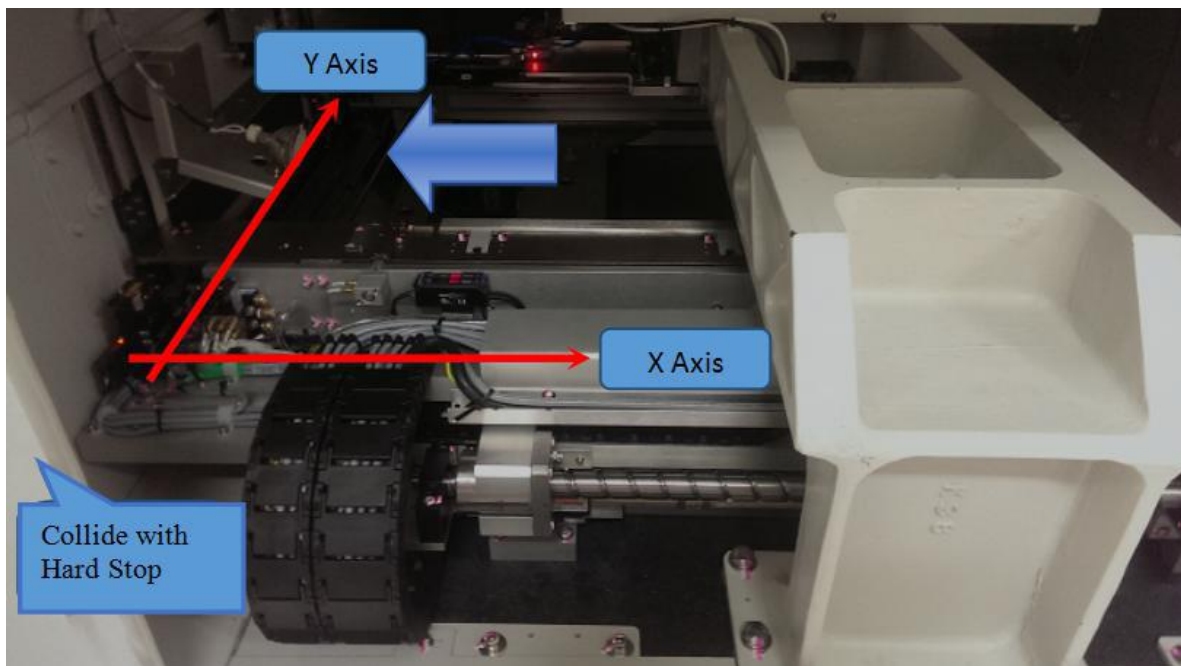


Figure 6 Minimum hard stop (V810 Std)

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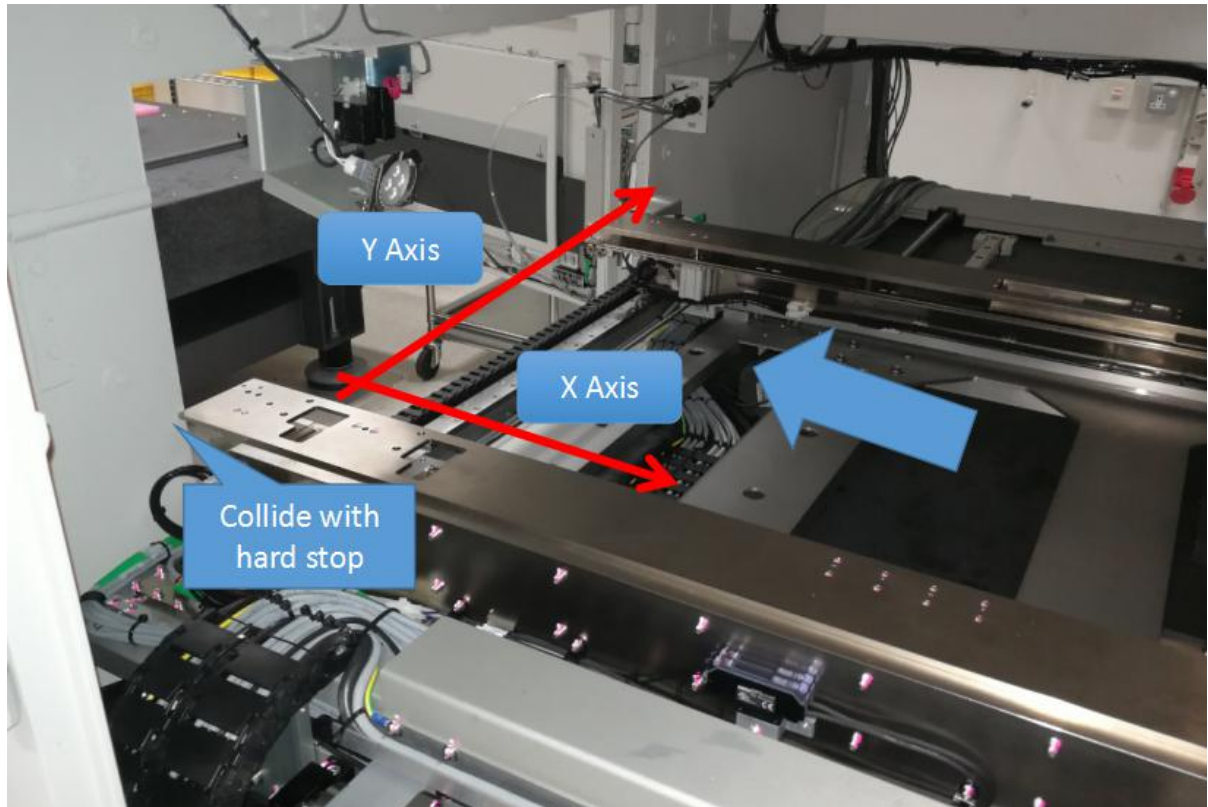


Figure 7 Minimum hard stop (V810 XXL)

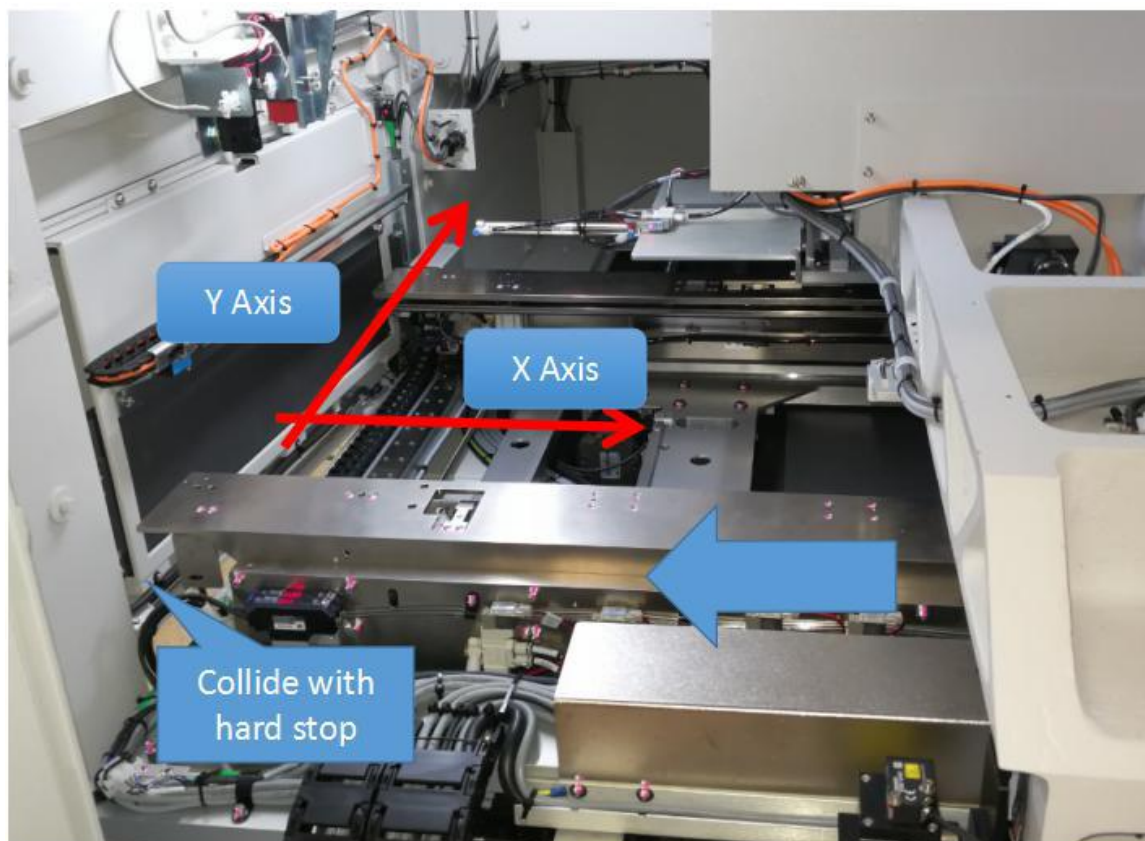


Figure 8 Minimum hard stop (V810 S2EX)

8. Reset all MS0, MS1 and MS2 at the MS Summary (Motion Supervisor Summary) as shown in Figure 9.
9. Once performed the reset for all the axis, the actual position for MS0, MS1 and MS2 will become zero as shown in Figure 10.

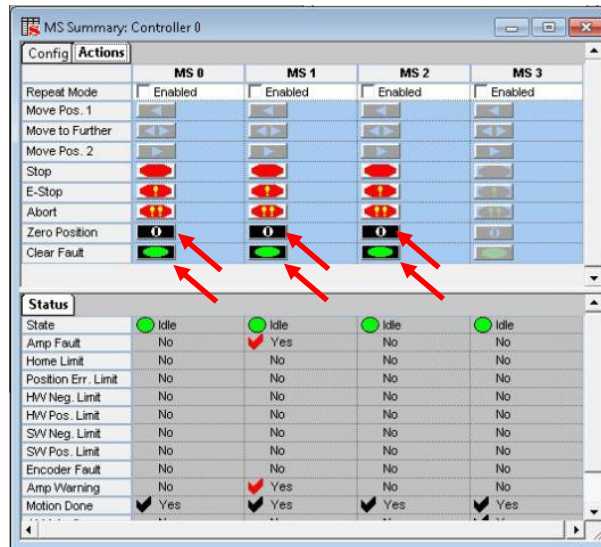


Figure 9 MS Summary (Motion Supervisor Summary)

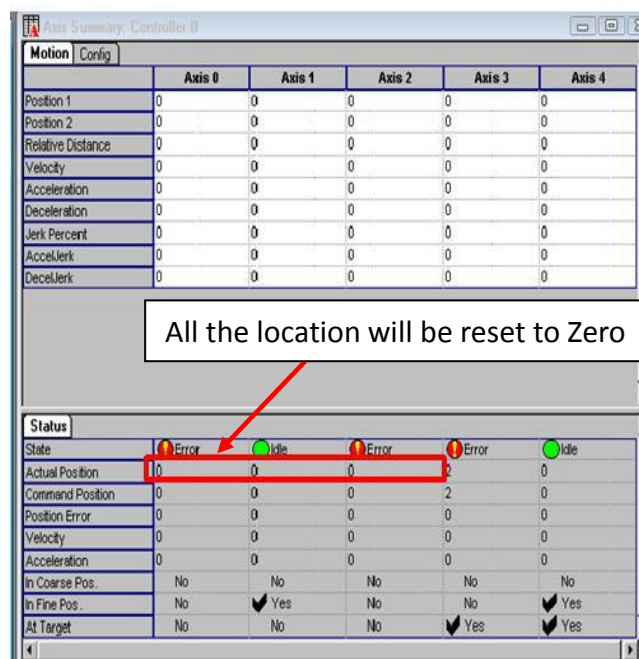


Figure 10 Axis Summary (Actual position)

9. Gently rotate the coupling for the X-Axis stage as shown in the Figure 11 until the Actual Position and Command Position in Figure 12 become 5990.

Remarks: Acceptable range for the Actual Position and Command Position will be from 5970~5990.



Figure 11 X-axis coupling screw

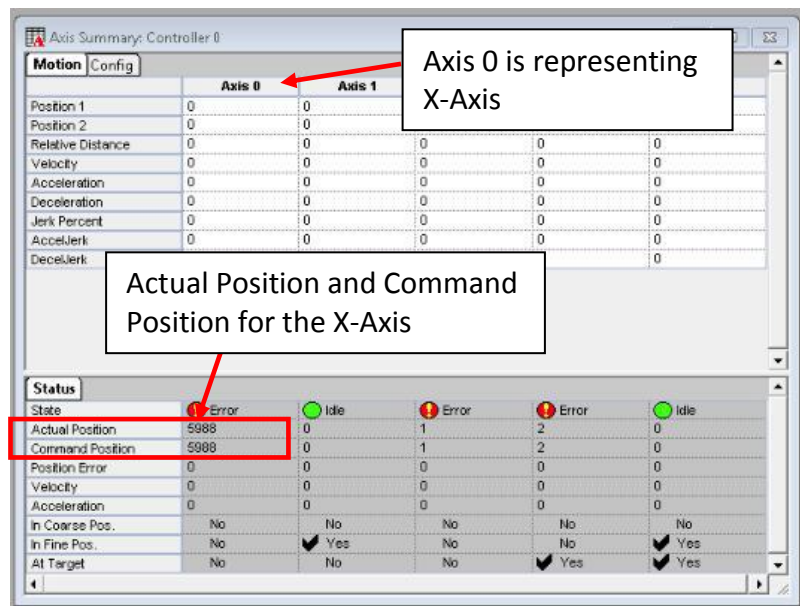


Figure 12 Actual Position and Command Position

10. Loosen the M3 screw of the limit switch as shown in Figure 13.

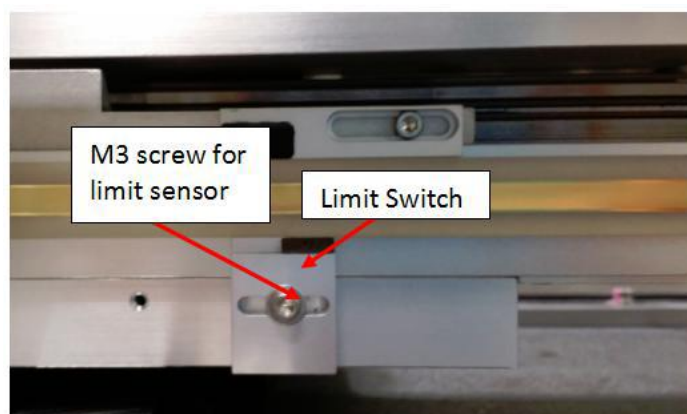


Figure 13 Limit Sensor for X-Axis

11. Adjust the limit switch shown in Figure 13 until the Home Sensor status become “High” at the Motor Summary as shown in Figure 14 and Figure 15.



Figure 14 Motor Summary (Home Sensor)

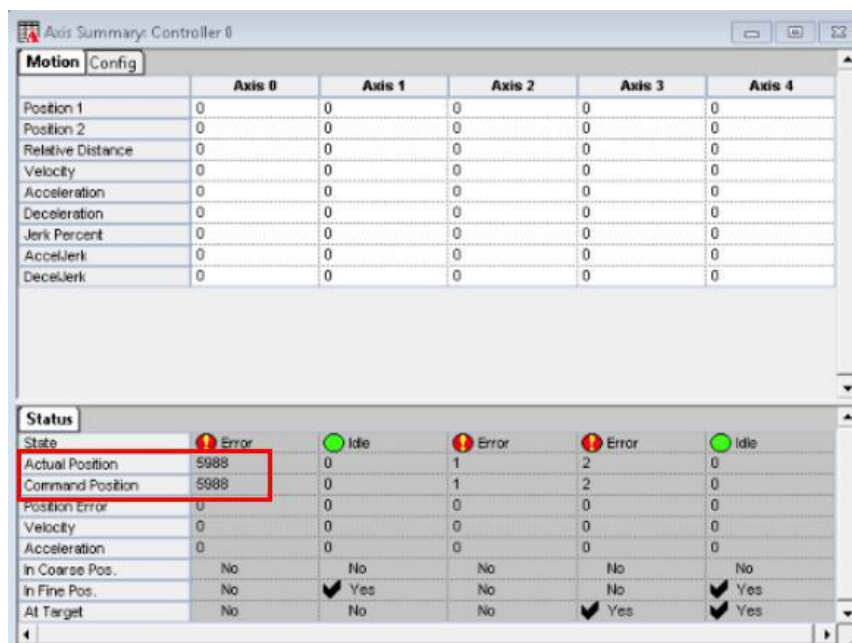


Figure 15 Axis Summary (Home Sensor)

12. Once the limit switch condition is acquired, tighten back the M3 screw for the limit sensor screw as shown in Figure 13.
13. Pull the X-Axis stage to the minimum hard stop.
14. Reset again the Axis 0 become zero as shown in Figure 16 and ensure the actual position for the Axis 0, Axis 1 and Axis 2 become zero as shown in Figure 17.

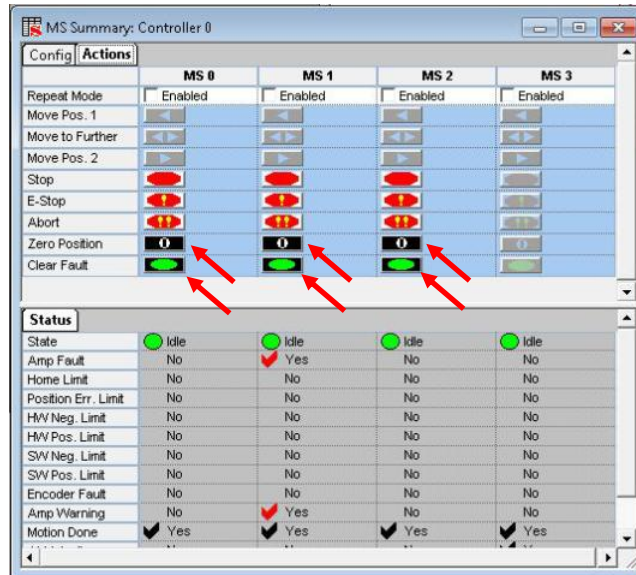


Figure 16 MS Summary

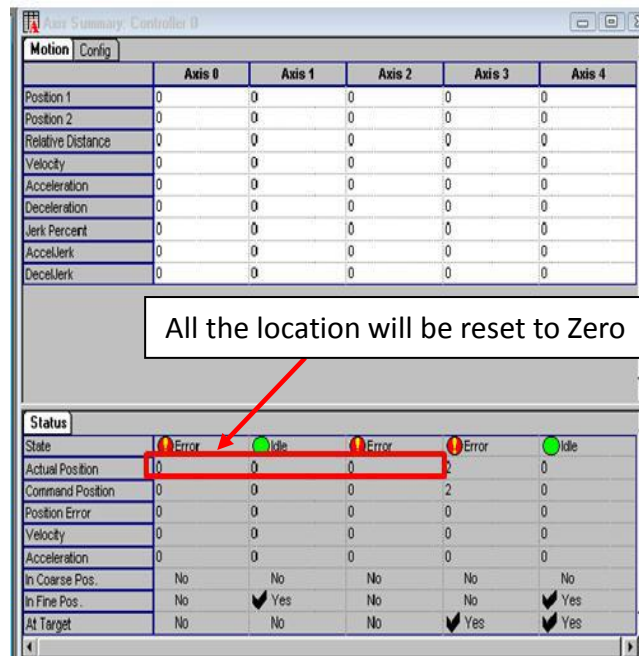


Figure 17 Axis Summary (Actual position)

15. Slowly rotate the X-axis coupling until see the status become “Low” at indicated at the “Home Sensor” as shown in Figure 18.

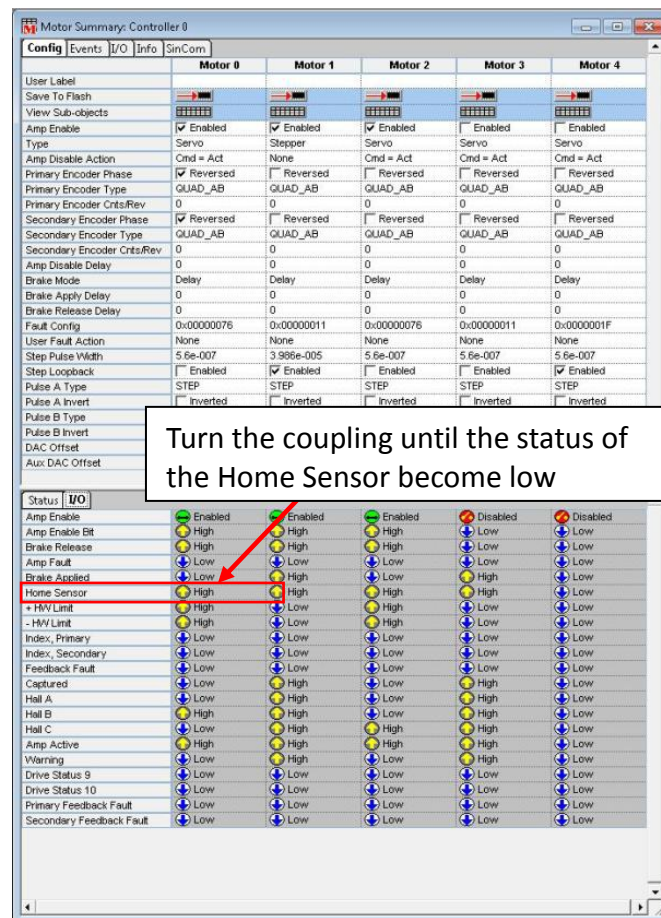
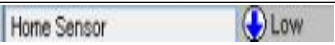
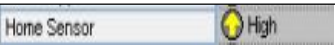


Figure 18 Motor Summary

16. Once the status becomes “Low”, X-axis stage should not move anymore.
17. Refer to the actual (From Axis Summary) and verify the limit sensor position. May refer to below table for counter reference.

X axis Actual position (Axis 0 at Axis summary)	Expected result at Motor 0 (Motor summary)
5000~5200	
5700~5990	

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18. Transition signal from low to high will happen at the range of 5000-6000 counts. If the result is not able to achieved as in above table (Low at 5000-5200 and High at 5700-5990), please do repeat from Step 5-17.
19. Once complete, exit the Motion Console software.
20. Push the XY-stage to the middle of the machine and close the front access door.
21. Open the V810 software. Go to the Services->Subsystem Status & Control-> Digital IO tab and Click Initialize as shown in Figure 19.

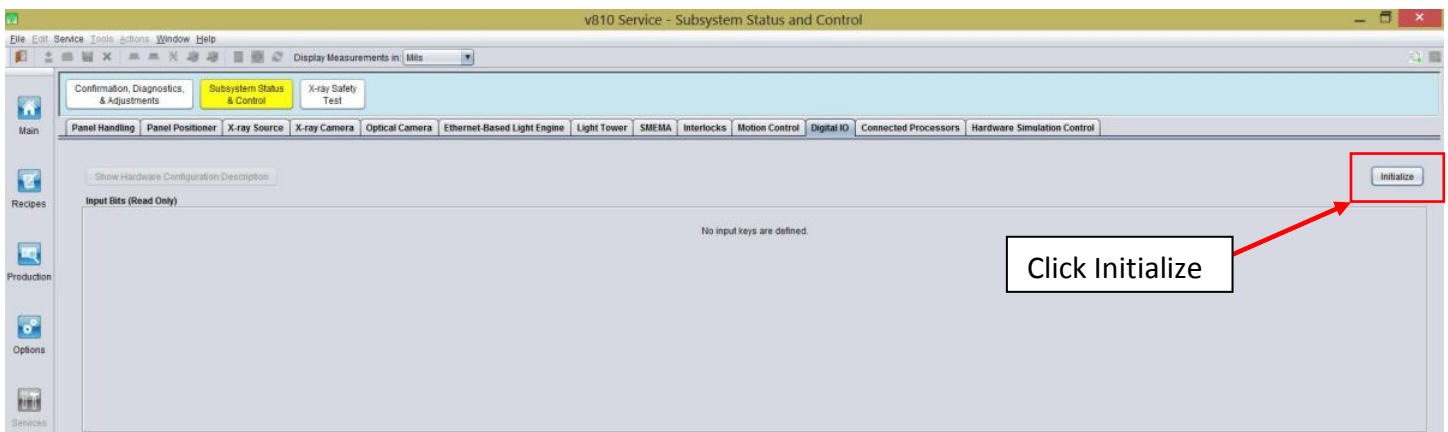


Figure 19 Digital IO

22. Go to the Panel Positioner tab and click Initialize. The limit sensor status for the X-axis should be clear. Disable the Motion for the X-axis and Y-axis as shown in Figure 20. Close the X-ray source and open the front door.

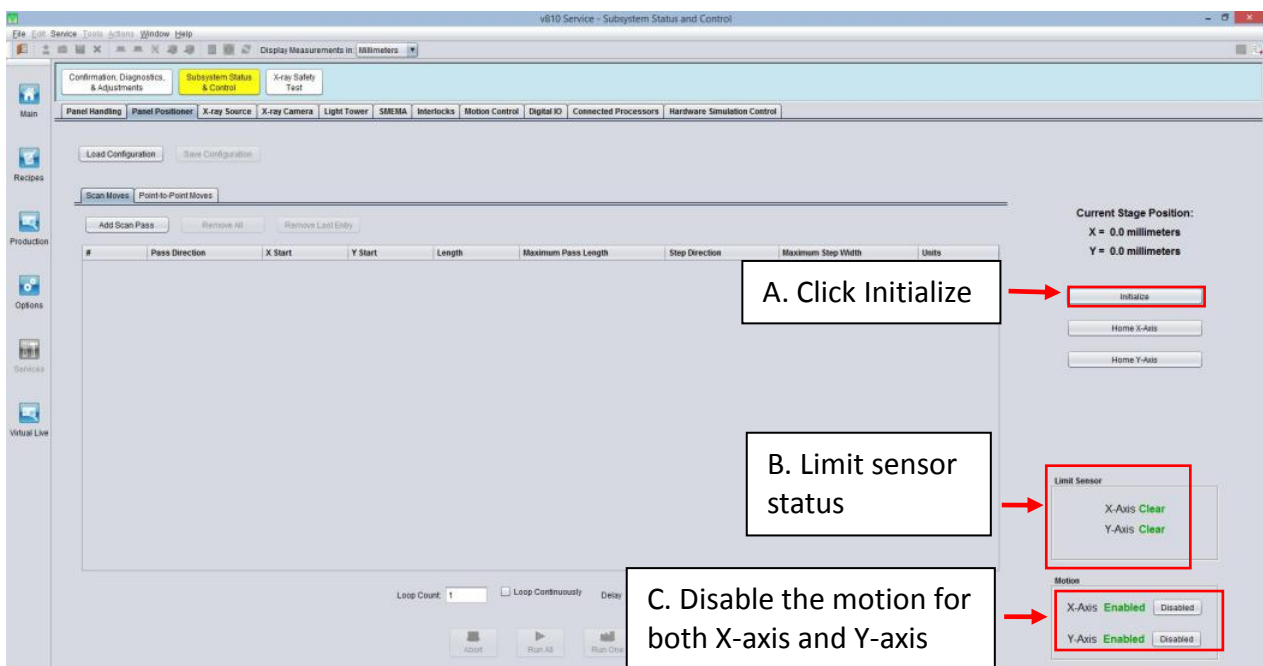


Figure 20 Panel Positioner

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23. Push the stage to the minimum hard stop (left outer barrier side) and status of the X-axis limit sensor should be indicated as “Blocked” as shown in Figure 21.

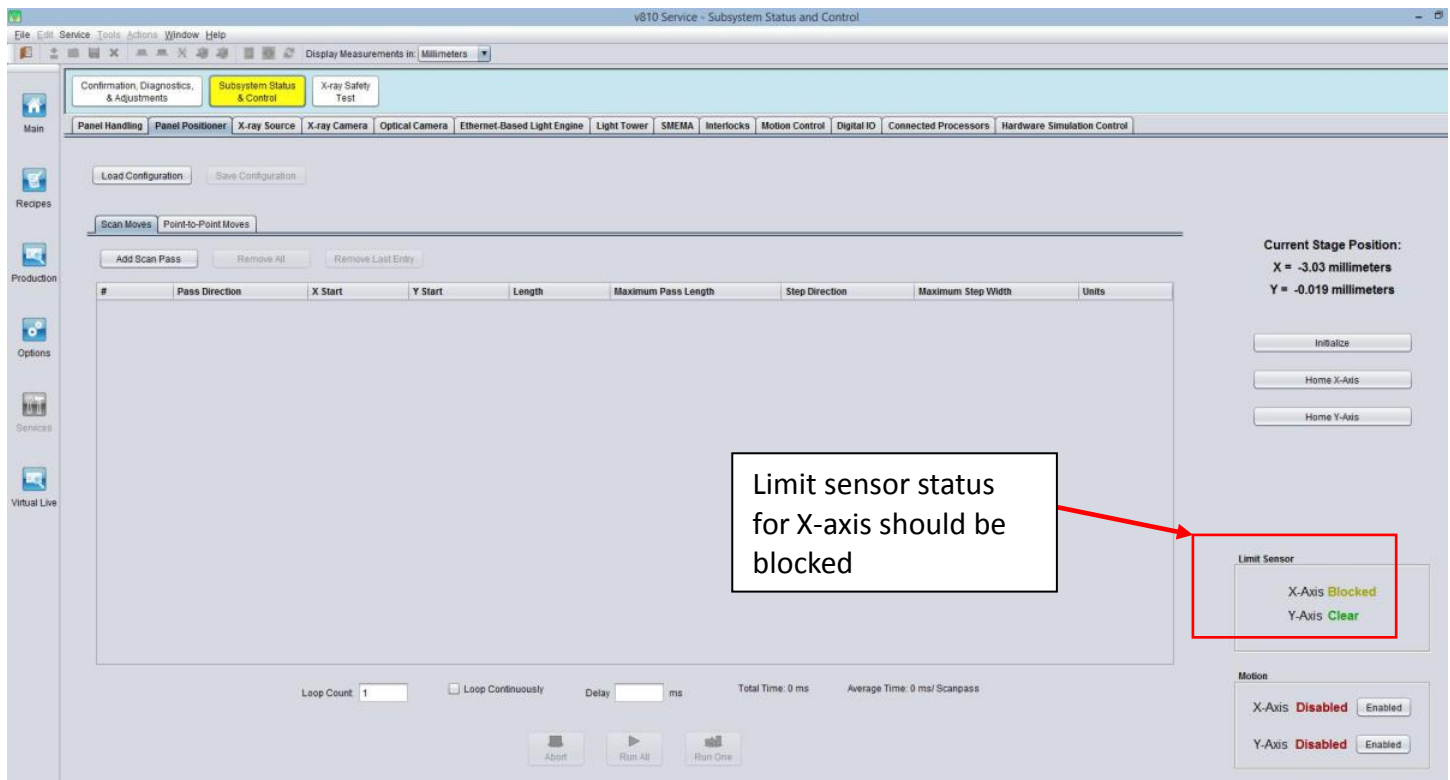


Figure 21 X-Axis Limit Sensor

24. If verification not be able to achieved, repeat Step 5-23.